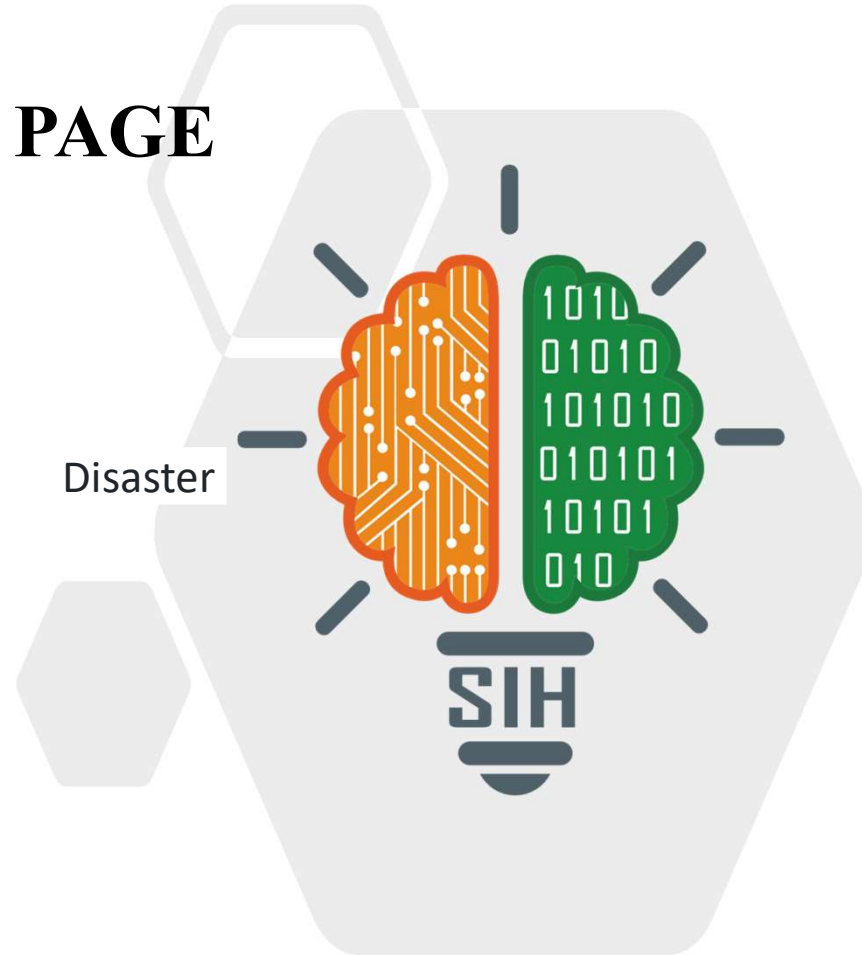


SMART INDIA HACKATHON 2024



TITLE PAGE

- **Problem Statement ID** –1687
- **Problem Statement Title**-Real Time Disaster Information Aggregation Software
- **Theme**-Disaster Management
- **PS Category**- Software
- **Team Name** - Quanta



Title: Real-Time Disaster Information Aggregation System



IDEA

- Develop a real-time disaster management system that aggregates and categorizes data from social media and other sources.
- **Functionality:**
 - ❖ **Text Classification:** Identifies the type of disaster (e.g., flood, earthquake) and assesses severity.
 - ❖ **Real-Time Analysis:** Processes live data to provide timely updates and alerts.
 - ❖ **Dashboard:** Offers a user-friendly interface for monitoring disaster status and receiving notifications.

Problem Resolution

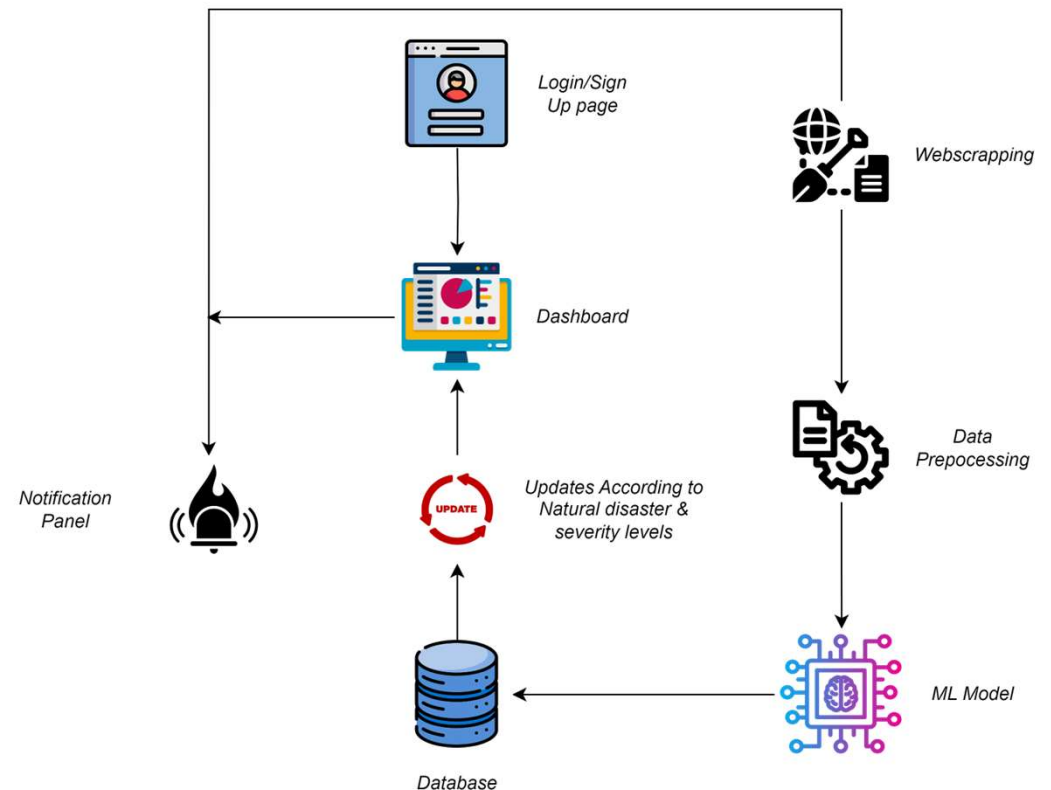
- **Data Overload:** Tackles large volumes of disaster-related text by using an efficient machine learning model for rapid classification.
- **Timely Information:** Provides real-time analysis and updates to support quicker disaster response.
- **Accuracy:** Leverages advanced text classification to improve disaster type and severity identification, minimizing false alerts.
- **Scalability:** Designed to scale with increasing data, ensuring long-term effectiveness as the system grows.

Unique Value Propositions

- **Enhanced Accuracy:** Utilizes BERT's bidirectional encoding for precise disaster type and severity classification.
- **Real-Time Processing:** Provides timely updates and alerts from live data.
- **Comprehensive Analysis:** Integrates news and social media data for a complete disaster overview.
- **User-Friendly Interface:** Simplifies monitoring and analysis with an easy-to-use interface.
- **Scalability and Adaptability:** Handles growing data and adapts to new disaster types effectively.
- **Why BERT?**
 - ❖ **Proven Performance:** Reliable across NLP tasks.
 - ❖ **Bidirectional Context:** Improves classification accuracy.
 - ❖ **Ease of Use:** Well-documented and straightforward to implement.
 - ❖ **Balanced Efficiency:** Strong accuracy with manageable resource needs.
 - ❖ **Cost-Effective:** Offers a solid trade-off between high performance and computational cost, ensuring good results without excessive resource expenditure.

Tools & Technologies Used:

- **Programming Languages:**
 - ❖ Python, HTML, CSS, JAVASCRIPT
- **Web Application Development:**
 - ❖ ReactJS, NodeJS, Express, Django are used to develop the whole Disaster Management Portal.
- **Algorithm Development:**
 - ❖ Transformer, PyTorch, Pandas & Datasets are the core technologies used for BERT model.
- **Cloud Database:**
 - ❖ MongoDB , a nonrelational database management system (DBMS)

Process Flow Architecture

FEASIBILITY AND VIABILITY



•Feasibility:

- ❖ Real-time data collection and processing using existing technologies.
- ❖ Integration of ML models with web frameworks for end-to-end solutions.

•Challenges:

- ❖ Handling large volumes of unstructured data.
- ❖ Ensuring the accuracy and relevance of the classified data.

•Strategies:

- ❖ Use distributed computing and optimized data pipelines.
- ❖ Continuous model evaluation and improvement through feedback loops.

•Potential Impact:

- ❖ **Faster Response:** Speeds up disaster analysis and decision-making, potentially saving lives.
- ❖ **Actionable Insights:** Helps response teams manage resources more effectively.
- ❖ **Improved Accuracy:** Reduces false alarms and enhances disaster identification.
- ❖ **Scalability:** Adapts to new disaster types and growing data over time.
- ❖ **Real-Time Monitoring:** Provides continuous updates for dynamic responses.
- ❖ **Resource Optimization:** Ensures efficient allocation of relief efforts.
- ❖ **Global Applicability:** Can be customized for different regions and disaster types.

• Benefits:

- ❖ **Social:** Enhances public safety and disaster preparedness with timely alerts and insights.
- ❖ **Economic:** Lowers disaster management costs by optimizing resources and preventing economic losses.
- ❖ **Environmental:** Supports quick actions to mitigate environmental damage and protect ecosystems.
- ❖ **Technological:** Uses advanced machine learning for accurate disaster classification and drives innovation in response systems.
- ❖ **Operational:** Improves response efficiency with real-time data, enhancing decision-making.
- ❖ **Computational Cost:** BERT offers high accuracy with moderate computational costs, making it scalable and cost effective.

RESEARCH AND REFERENCES



- Dataset : <https://www.kaggle.com/datasets/vstepanenko/disaster-tweets>
- Research Paper: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3610676
- Article: <https://www.undp.org/belarus/stories/emergency-communications-power-social-media>